

ALEX CHEVRIER

Software Engineer — Distributed Systems & Low-Latency Infrastructure
Hong Kong | +852 55189053 | chevrier.alex@gmail.com
linkedin.com/in/alex-chevrier-323365b4 | github.com/alchevrier

SKILLS

Languages: C++23, Java 25, Kotlin, TypeScript, SQL
Low-latency: mmap, lock-free SPSC, memory ordering, perf/massif, JMH/JFR,
MemorySegment / Panama API, Google Benchmark, CMake, Conan
Systems: Kafka, binary TCP/NIO, Raft consensus, Docker, Kubernetes, AWS

EXPERIENCE

Morgan Stanley, Hong Kong — Data Service Developer (Contract) Nov 2025–Present

- Designed greenfield messaging service from scratch: reverse-engineered undocumented Python batch system, implemented streaming pipeline with Spring Kafka + Java 25.
- Maintain production Kafka infrastructure across isolated data centres for cross-region financial data consistency.

We+, Hong Kong — Java Developer Sept 2023–Nov 2025

- Migrated the highest number of microservices to Java 21 across a major French investment bank's FX platform.
- Rebuilt RHEL9 test environments post team transfer; produced end-to-end runbooks covering load balancers, SSL, JKS — eliminated external team dependency.

LandyTech, London — Software Engineer Oct 2019–Aug 2023

- Contributed to monolith-to-microservices migration and automated risk data pipeline.

Redbite Solutions, Cambridge — Software Engineer Nov 2016–Oct 2019

- Built web application from scratch; designed and implemented OAuth2 with Spring Security after identifying missing authentication layer.

PROJECTS

Low-Latency Log Engine (C++23) | github.com/alchevrier/low-latency-log-engine
mmap-backed append-only log with lock-free SPSC ring buffer. 16.1 ns per 64-byte append (3.70 GB/s). perf: 68% backend-bound — store bandwidth to L1D, no algorithmic waste. massif: zero heap allocation across 1M iterations. Compile-time hardware contracts via C++23 concepts. ADR-driven design.

Mini-Kafka (Java 25) | github.com/alchevrier/distributed-messaging
Kafka-like distributed log: append-only storage, binary TCP/NIO, Raft consensus. Eliminated 655 ms GC pauses via off-heap MemorySegment slabs (p99.99 –77%, JFR-confirmed).

EDUCATION

ENSEIRB-MATMECA, Bordeaux — M.Sc. Computer Science 2013–2016
Admitted by competitive national examination (Grande École entrance).

